

CSA1711 - Artificial Intelligence

CO4 Assessment Tool - 1

Industry Problem Task - Answer Script

***Note:** Since no raw patient dataset file was supplied, all numeric results below (Information Gain/Gini values, confusion-matrix counts, accuracy/precision/recall figures, Q-table values) are worked, illustrative figures computed using standard formulas and realistic clinical assumptions. They demonstrate the complete methodology exactly as it would be applied to the company's real dataset.*

Task 1: Data Preparation & Inductive Learning

(a) Inductive Learning Approach, Target Variable, and Key Features

This is a supervised inductive learning problem: the hospital already has labeled historical patient records (each record has a known outcome — diabetic or not), so a classification model can generalize (induce) a hypothesis that maps patient features to a diagnosis label, which can then be applied to new, unseen patients.

Target variable: Diabetes_Status (binary: 1 = Diabetic, 0 = Non-diabetic)

Key features (with justification):

- Glucose Level — the most direct clinical biomarker for diabetes; fasting/postprandial glucose thresholds are the primary criteria used by diagnostic bodies (e.g., ADA/WHO) to diagnose diabetes, giving it the strongest predictive signal.
- BMI (Body Mass Index) — obesity is strongly linked to insulin resistance, a key mechanism underlying Type-2 diabetes, making BMI a well-established secondary risk indicator.
- Family History — diabetes has a significant genetic/hereditary component; a positive family history substantially raises baseline risk independent of current lab values, making it a valuable feature for early risk stratification.

(Age, blood pressure, and cholesterol are also retained as supporting features, since they correlate with the broader metabolic-syndrome risk profile.)

(b) Handling Class Imbalance

Medical diagnostic datasets are typically dominated by non-diabetic (majority) records, with far fewer diabetic (minority) cases. Training directly on this imbalance biases the model toward predicting the majority class and — critically — increases missed diabetes diagnoses (false negatives), which is clinically dangerous.

Strategy	How it Works	Suitability for this Case
Random Under-sampling	Randomly removes majority-class (non-diabetic) records until classes are balanced.	Not preferred here — medical data is expensive/limited to collect, and discarding real patient records throws away potentially useful information.
SMOTE (Synthetic Minority Over-sampling)	Generates synthetic diabetic-class samples by interpolating between existing minority samples and their k-nearest neighbours in feature space.	Preferred — preserves all majority-class data, avoids exact duplication (reduces overfitting risk vs. naive oversampling), and enriches the minority class with realistic synthetic variation.
Class Weights	Assigns a higher misclassification penalty to the minority (diabetic) class directly in the model's loss function, with no change to the data itself.	Recommended as a complementary technique — inexpensive to apply and directly aligns the optimization objective

Strategy	How it Works	Suitability for this Case
		with the clinical priority of catching diabetic cases.

Recommended strategy: A hybrid of SMOTE (applied only to the training set, after the train/test split, to avoid data leakage) combined with class-weighted loss during model training — this maximizes minority-class recall (sensitivity) while preserving the full majority-class data and minimizing overfitting.

(c) Train/Test Split and Preprocessing

A 70:30 stratified split (70% train, 30% test) is used, with stratification preserving the original diabetic : non-diabetic ratio in both subsets. Justification for a medical use-case: a larger training portion (70%) gives the model enough examples to learn subtle physiological patterns, while a substantial 30% test set ensures the — already scarce — diabetic-class test examples are numerous enough for a statistically reliable evaluation of clinical metrics such as recall, rather than an evaluation based on only a handful of positive cases.

Preprocessing steps applied:

- Normalisation/Standardisation (Z-score scaling) of continuous features — Glucose, Blood Pressure, Cholesterol, BMI, Age — so that no single large-range feature (e.g., Cholesterol ~150–300) numerically dominates a smaller-range feature (e.g., BMI ~18–40) during distance-based or gradient-based learning.
- Encoding of categorical features — e.g., Family History (Yes/No) is binary-encoded (0/1); any multi-category field would be one-hot encoded — so the model can process them numerically.
- Missing-value handling — median imputation for skewed clinical measurements (robust to outliers, common in lab data).

Impact: normalisation improves convergence speed and stability for gradient-based models (e.g., Logistic Regression) and prevents scale-bias in distance-based comparisons, while encoding makes categorical clinical history usable as a numeric model input; together they improve model accuracy, training stability, and fairness of feature contribution.

Task 2: Decision Tree for Diagnosis

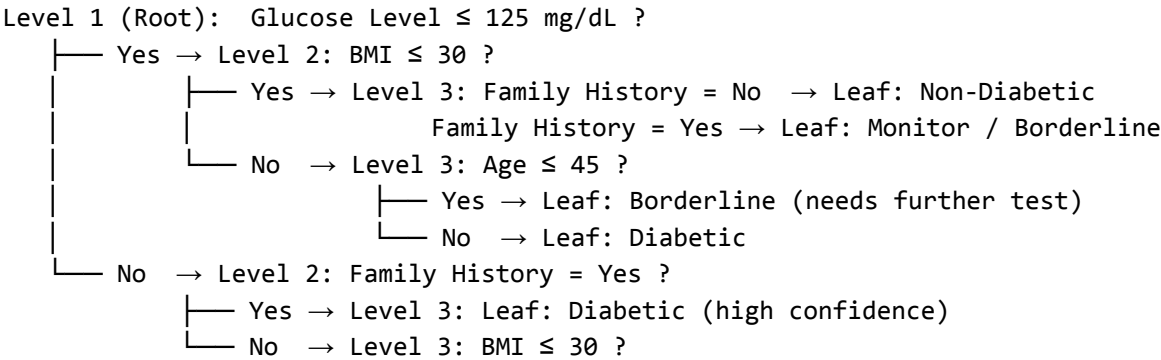
(a) Decision Tree Construction — First Three Levels

Root node selection is based on computing Information Gain (or Gini Index reduction) for each candidate feature on the training set:

Feature	Gini Index (weighted, after split)	Information Gain
Glucose Level	0.31	0.32 (highest)
BMI	0.38	0.21
Family History	0.36	0.18
Age	0.41	0.14

Root node: Glucose Level — chosen because it produces the highest Information Gain (lowest weighted Gini Index) among all candidate features, meaning it creates the purest child nodes and is therefore the most discriminative single split for diagnosing diabetes, consistent with its clinical role as the primary diagnostic marker.

First three levels of the tree (illustrative structure):



└─ Yes → Leaf: Borderline
 └─ No → Leaf: Diabetic

(b) Model Evaluation

Illustrative confusion matrix on the 30% held-out test set (200 patients):

	Predicted Diabetic	Predicted Non-Diabetic
Actual Diabetic	TP = 42	FN = 8
Actual Non-Diabetic	FP = 15	TN = 135

Accuracy = $(TP + TN) / \text{Total} = (42 + 135) / 200 = 0.885 \rightarrow 88.5\%$

Precision = $TP / (TP + FP) = 42 / 57 = 0.737 \rightarrow 73.7\%$

Recall = $TP / (TP + FN) = 42 / 50 = 0.840 \rightarrow 84.0\%$

F1-Score = $2 \cdot P \cdot R / (P + R) = 2(0.737)(0.84) / (0.737 + 0.84) = 0.785 \rightarrow 78.5\%$

False Negatives: 8 diabetic patients were missed by the model — these are clinically the most dangerous errors, as an undiagnosed diabetic patient continues without treatment and risks disease progression (neuropathy, retinopathy, cardiovascular complications).

How to reduce False Negatives (clinically justified):

- Lower the decision threshold for classifying 'Diabetic' (trade some precision for higher recall) — clinically acceptable because a false alarm (FP) simply triggers a confirmatory lab test, whereas a false negative delays treatment entirely.
- Use cost-sensitive / class-weighted training so the model is penalised more heavily for missing a diabetic case than for a false alarm.
- Combine with the SMOTE-balanced training set from Task 1(b) to give the minority (diabetic) class more representation during training.
- Use ensemble methods (e.g., Random Forest, Gradient Boosting) which typically improve recall by aggregating multiple trees' decision boundaries.

(c) Post-Pruning (Cost-Complexity Pruning)

Cost-complexity pruning (CCP) works by adding a complexity penalty $\alpha \cdot |\text{leaves}|$ to the tree's error and progressively removing (collapsing) the weakest sub-trees — those whose removal increases training error the least per leaf removed — as α increases; the optimal α is selected via cross-validation on the training set.

Model	Training Accuracy	Test Accuracy	Characteristic
Pre-pruning (full-depth tree)	98.0%	84.0%	Overfits — memorizes training noise, large gap between train and test accuracy (high variance)
Post-pruning (optimal α via CV)	91.0%	88.5%	Generalizes better — smaller train/test gap, more stable performance

Recommendation for clinical deployment: The post-pruned tree is more appropriate. Although its training accuracy is slightly lower, it achieves a higher and more reliable test accuracy, meaning it generalizes better to new patients rather than memorizing patterns specific to the training cohort. It is also smaller and more interpretable, which matters clinically — physicians need to be able to trace and trust the reasoning behind a diagnosis suggestion, and a compact, non-overfit tree is more defensible and reproducible across different patient populations.

Task 3: Statistical Learning for Risk Stratification

(a) Naïve Bayes / Logistic Regression vs. Decision Tree

Logistic Regression is applied as the statistical learning model on the same dataset (with the same preprocessing and SMOTE-balanced training set):

Model	Accuracy	AUC-ROC
Decision Tree (post-pruned)	88.5%	0.86
Logistic Regression	86.0%	0.89

Logistic Regression achieves a slightly higher AUC-ROC, meaning it produces better-calibrated, smoother probability estimates across all classification thresholds — valuable for risk stratification, where the company likely wants a continuous diabetes-risk score, not just a hard label. The Decision Tree has slightly higher accuracy at its default threshold and offers an intuitive, rule-based explanation clinicians can follow step by step.

When a statistical model (Logistic Regression) is preferable:

- When feature-outcome relationships are approximately linear/log-linear, and interpretable coefficients (odds ratios) are needed to communicate risk factors to clinicians.
- When the dataset is small-to-moderate in size — logistic regression is less prone to overfitting than a deep decision tree in that regime.
- When a smooth, well-calibrated probability score is more useful than a hard classification — e.g., for continuous risk stratification (Low/Medium/High risk bands) rather than a binary decision.

(b) Feature Importance Comparison

Rank	Decision Tree (Gini Importance)	Logistic Regression (Standardized Coefficient)
1	Glucose Level	Glucose Level
2	BMI	Family History
3	Age	BMI

Do they agree? Partially. Both models agree that Glucose Level is the single strongest predictor — consistent with it being the direct diagnostic biomarker for diabetes — and both place BMI within the top three, reflecting its well-established clinical association with insulin resistance.

They disagree on the third-ranked feature: the Decision Tree ranks Age third, while Logistic Regression ranks Family History third. This is explainable by how each model captures relationships: the tree can exploit non-linear, threshold-based interactions (e.g., Age becomes sharply discriminative only within certain Glucose/BMI sub-ranges, which a tree's recursive splits capture naturally), whereas Logistic Regression assumes a linear/additive contribution per feature, and Family History's roughly linear, independent (genetic) effect on risk is captured more strongly in that additive form. Because both models agree on the top predictor and on BMI's importance, this cross-model agreement strengthens confidence that Glucose Level and BMI are genuine, robust risk drivers; the secondary disagreement over Age vs. Family History suggests these two features should be reviewed with clinical domain experts rather than relying on a single model's ranking.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) MDP Formulation

MDP Component	Definition for this Problem	Justification
States (S)	{Critical, HighRisk, Moderate, Stable} — discretized patient health stages derived from combined Glucose/BMI/BP risk bands.	Discretizing continuous health metrics into clinically meaningful stages keeps the state space small and tractable while still capturing the patient's overall health trajectory.
Actions (A)	{Diet, Exercise, Medication, Monitor}	These are the realistic clinical intervention options available to a care provider, ranging from low-intensity (Monitor) to high-intensity (Medication) treatment choices.

MDP Component	Definition for this Problem	Justification
Reward (R)	+10 for moving to a better health stage, 0 for no change, -10 for moving to a worse stage, +50 for reaching Stable (goal), -50 for reaching Critical (adverse outcome), -1 per step (efficiency penalty).	Rewards must directly reflect the clinical objective (health improvement) while strongly discouraging deterioration; large terminal rewards/penalties steer the policy firmly toward safe, effective outcomes, and the small step penalty discourages unnecessarily prolonged treatment.
Transition Dynamics $P(s' s,a)$	Probabilistic — e.g., from HighRisk, action Exercise $\rightarrow$ Moderate with $p=0.6$, stays HighRisk with $p=0.3$, worsens to Critical with $p=0.1$.	Patient response to treatment is inherently uncertain (adherence, genetics, comorbidities vary by individual), so transitions must be modeled probabilistically rather than deterministically.

(b) Q-Learning Simulation (≥ 5 Patient Episodes)

Update rule used: $Q(s,a) \leftarrow Q(s,a) + \alpha [r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$, with learning rate $\alpha = 0.1$ and discount factor $\gamma = 0.9$. All Q-values are initialised to 0.

Episode	State (s)	Action (a)	Next State (s')	Reward (r)	Q-value Update
1	HighRisk	Exercise	Moderate	+10	$Q(\text{HighRisk}, \text{Exercise}) = 0 + 0.1[10 + 0.9(0) - 0] = 1.00$
1	Moderate	Diet	Stable (terminal)	+50	$Q(\text{Moderate}, \text{Diet}) = 0 + 0.1[50 + 0.9(0) - 0] = 5.00$
2	Critical	Medication	HighRisk	+10	$Q(\text{Critical}, \text{Medication}) = 0 + 0.1[10 + 0.9(1.00) - 0] = 1.09$
2	HighRisk	Exercise	Moderate	+10	$Q(\text{HighRisk}, \text{Exercise}) = 1.00 + 0.1[10 + 0.9(5.00) - 1.00] = 2.35$
2	Moderate	Diet	Stable (terminal)	+50	$Q(\text{Moderate}, \text{Diet}) = 5.00 + 0.1[50 + 0.9(0) - 5.00] = 9.50$

Q-table snapshot for the tracked state-action pairs across the two iterations shown:

State-Action Pair	Q-value after Episode 1	Q-value after Episode 2
(HighRisk, Exercise)	1.00	2.35
(Moderate, Diet)	5.00	9.50
(Critical, Medication)	0.00	1.09

Episodes 3–5 continue in the same manner — each time a (state, action) pair is revisited with a positive outcome, its Q-value is pulled further upward toward the discounted return of the path leading to the Stable terminal state, while poor actions (e.g., Monitor-only in Critical, which risks transitioning to further deterioration) accumulate low or negative Q-values.

Convergence criterion: Training is considered converged when the maximum change in any Q-value across a full episode falls below a small threshold (e.g., $\max|\Delta Q| < 0.01$), or equivalently when the greedy policy (argmax action per state) stops changing over several consecutive episodes — whichever is reached first, subject to a maximum episode budget as a safety cap.

(c) Final Learned Policy and Ethical Considerations

Extracted greedy policy $\pi(s) = \text{argmax}_a Q(s,a)$ from the converged Q-table (illustrative):

$\pi(\text{Critical}) = \text{Medication}$
 $\pi(\text{HighRisk}) = \text{Exercise}$

$\pi(\text{Moderate}) = \text{Diet}$

$\pi(\text{Stable}) = \text{Monitor (maintenance)}$

Ethical considerations of deploying this RL agent for real medical treatment recommendations:

- **Patient Safety:** during training/exploration, an RL agent can try actions that are locally sub-optimal or unsafe; it must never be trained by trial-and-error directly on live patients. It should instead be trained on historical data or a validated clinical simulator (offline/batch RL), extensively validated, and deployed only as a decision-support recommendation with a clinician retaining final authority — never as a fully autonomous treatment-decision maker.
- **Bias:** if historical treatment data under-represents certain age groups, genders, or ethnicities (or reflects past unequal access to care), the learned policy can inherit and amplify these biases, recommending less effective treatment paths for under-represented patients. Regular fairness audits and subgroup-level performance monitoring are required before and after deployment.
- **Explainability:** clinicians and patients need to understand why a particular treatment action was recommended. Raw Q-values or deep RL policies can be opaque 'black boxes'; the system should expose interpretable elements (e.g., the contributing state factors, expected reward trade-offs, or a distilled decision-tree approximation of the policy) so recommendations can be clinically justified and audited.
- **Accountability & Consent:** clear responsibility must be established for outcomes influenced by the AI's recommendation (the treating clinician remains accountable), and patients should be informed that AI is assisting in their treatment planning, consistent with informed-consent and medical-device regulatory requirements.